

FROBENIUS MANIFOLDS AND QUANTUM COHOMOLOGY

The aim is to define Frobenius manifolds and sketch the example of quantum cohomology. In general, examples arise from non-commutative Hodge theory (deformations of Calabi-Yau A^∞ categories)¹. There are then two types coming from deformations of Fukaya categories (A -model) or derived categories (B -model), and these are related by mirror symmetry. Quantum cohomology is about deformations of Fukaya categories. Taking derived categories of matrix factorisations relates to Kyoji Saito's theory of primitive forms in singularity theory.

Remarks 0.1. (1) We will work over $\mathbb{C}$ so our Frobenius manifolds will be complex manifolds. You could work over $\mathbb{R}$ and have smooth manifolds but no-one ever seems to.
 (2) For a general variety $H^*(X, \mathbb{C})$ is only graded commutative. This leads to Frobenius supermanifolds. We won't go there: we just consider even cohomology.

1. FROBENIUS ALGEBRAS

Definition 1.1. A (commutative, associative, unital) Frobenius algebra is a commutative algebra A with a non-degenerate symmetric bilinear form $g: A \otimes A \rightarrow \mathbb{C}$ such that $g(u * v, w) = g(u, v * w)$ for all $u, v, w \in A$.

Remarks 1.2. (1) All Frobenius algebras will be assumed commutative, associative, unital.
 (2) Given this, we can equivalently define a Frobenius algebra as a commutative algebra A with a linear form $\theta: A \rightarrow \mathbb{C}$ such that the symmetric bilinear form $g(u, v) = \theta(u * v)$ is non-degenerate. Indeed, given g as in Definition 1.1 we can take $\theta(u) = g(u, e)$, where $e \in A$ is the identity.
 (3) Non-degenerate means that the induced map $A \rightarrow A^*$ is an isomorphism. This implies that A is finite-dimensional.
 (4) There is a famous old theorem that a (1+1) dimensional TQFT is equivalent data to a Frobenius algebra.²

Example 1.3. Let X be a compact complex manifold and set $A = H^{2*}(X, \mathbb{C})$. Then the cup product makes A into an associative, commutative algebra, and Poincaré duality says that we can take $\theta: A \rightarrow \mathbb{C}$ to be pairing with the fundamental class. Thus

$$g(\phi_1, \phi_2) = \int_X \phi_1 \wedge \phi_2.$$

¹See Barannikov and Kontsevich, Frobenius manifolds and formality of Lie algebras of polyvector fields.

²See e.g. Atiyah, An introduction to topological quantum field theories.

2. ASSOCIATIVITY CONDITION

Take a basis $A = \bigoplus_{i=1}^n \mathbb{C}e_i$, with $e_1 = e$, and set

$$c_{ijk} = g(e_i * e_j, e_k), \quad g_{jk} = g(e_j, e_k) = c_{1jk}.$$

Then c_{ijk} is totally symmetric and g_{jk} is non-degenerate. Conversely given c_{ijk} satisfying these conditions, the only extra constraint is associativity.

Lemma 2.1. *Associativity is equivalent to*

$$\sum_{r,s} c_{ijr} \cdot g^{rs} \cdot c_{kls} = \sum_{r,s} c_{jkr} \cdot g^{rs} \cdot c_{ils} \quad \forall i, j, k, l,$$

where g^{ij} denotes the inverse matrix to g_{ij} .

Proof. Write $e_i * e_j = \sum_k m_{ij}^k e_k$. Then associativity is

$$\sum_p m_{ij}^p \cdot m_{pk}^q = \sum_p m_{jk}^p \cdot m_{pi}^q, \quad \forall i, j, k, q. \quad (1)$$

But by definition $c_{ijk} = \sum_r m_{ij}^r \cdot g_{rk}$ which implies $m_{ij}^k = \sum_r c_{ijr} \cdot g^{rk}$. Substituting this into (1) gives

$$\sum_{p,q,s} c_{ijr} \cdot g^{rp} \cdot c_{pks} \cdot g^{sq} = \sum_{p,r,s} c_{jkr} \cdot g^{rp} \cdot c_{pis} \cdot g^{sq}.$$

Multiplying both sides by g_{ql} , summing over q , and replacing p with s gives the result. $\square$

3. FROBENIUS MANIFOLDS

To give a Frobenius structure on a manifold is roughly to give the structure of a Frobenius algebra on each tangent space. But there are integrability conditions.

Definition 3.1. A Frobenius structure on a complex manifold M consists of

- (i) a metric: a holomorphic map $g: T_M \otimes T_M \rightarrow \mathcal{O}_M$, inducing a symmetric and non-degenerate bilinear form on each tangent space $T_{M,p}$,
- (ii) a product: a holomorphic map $*$: $T_M \otimes T_M \rightarrow T_M$, inducing on each tangent space $T_{M,p}$ the structure of a commutative and associative algebra,

satisfying the following three conditions:

- (F1) the Levi-Civita connection ∇ on T_M associated to the metric g is flat,
- (F2) there exists a flat vector field $e: \mathcal{O}_M \rightarrow T_M$ which is a unit for the product,
- (F3) in local flat co-ordinates $(t_1, \dots, t_n)$ for the metric g there exists a locally-defined function $F(t_1, \dots, t_n)$ such that

$$g\left(\frac{\partial}{\partial t_i} * \frac{\partial}{\partial t_j}, \frac{\partial}{\partial t_k}\right) = \frac{\partial^3 F}{\partial t_i \partial t_j \partial t_k} = g\left(\frac{\partial}{\partial t_i}, \frac{\partial}{\partial t_j} * \frac{\partial}{\partial t_k}\right). \quad (2)$$

By a Frobenius manifold we of course mean a manifold equipped with a Frobenius structure. The complex dimension of the underlying manifold M will usually be denoted n .

- Remarks 3.2.** (1) As part of the definition of a Frobenius manifold one also usually imposes the existence of an Euler vector field. We leave that for later.
- (2) The vector field e of (F2) is clearly unique: it is called the identity vector field.
- (3) The function F in condition (F3) is called a prepotential: on any given patch it is well-defined up to the addition of quadratic polynomials in the flat co-ordinates t_i .

4. IN FLAT CO-ORDINATES

We often take local flat co-ordinates $(t_1, \dots, t_n)$ as in (F3). We can assume $e = \frac{\partial}{\partial t_1}$. The Frobenius structure is then determined by the prepotential F . The form

$$g_{ij} = g\left(\frac{\partial}{\partial t_i}, \frac{\partial}{\partial t_j}\right) = \frac{\partial^3 F}{\partial t_1 \partial t_i \partial t_j}$$

must be non-degenerate and constant. The only other constraint is then associativity, which gives the famous WDVV equations:

$$\sum_{p,s} \frac{\partial^3 \mathcal{F}}{\partial t_i \partial t_j \partial t_r} \cdot g^{rs} \cdot \frac{\partial^3 \mathcal{F}}{\partial t_k \partial t_l \partial t_s} = \sum_{p,s} \frac{\partial^3 \mathcal{F}}{\partial t_j \partial t_k \partial t_r} \cdot g^{rs} \cdot \frac{\partial^3 \mathcal{F}}{\partial t_i \partial t_l \partial t_s}, \quad \forall i, j, k, l.$$

Example 4.1. Let A be a Frobenius algebra. Take M to be the underlying vector space of A viewed as a complex manifold. The tangent space at any point of M is naturally identified with A so we get a ‘constant’ Frobenius structure on M .

Take a basis $e_1, \dots, e_n$ for A with $e_1 = e$. An element of M is then of the form $\sum_i t_i e_i$ with $t_i \in \mathbb{C}$ and we take the t_i as flat co-ordinates. Under the identification between the tangent space of M and the vector space A , the element $\frac{\partial}{\partial t_i}$ corresponds to e_i . The prepotential is the cubic

$$\mathcal{F} = \frac{1}{3!} \sum_{i,j,k} c_{ijk} t_i t_j t_k, \quad c_{ijk} = g(e_i * e_j, e_k) \in \mathbb{C}.$$

5. GROMOV-WITTEN INVARIANTS

Let X be a smooth projective variety. Gromov-Witten theory gives maps

$$I_{g,n,\beta}: H^*(X, \mathbb{C})^{\otimes n} \rightarrow H^*(\bar{M}_{g,n}, \mathbb{C}),$$

for $g, n \geq 0$ in the stable range, and $\beta \in H_2(X, \mathbb{Z})$. Stable range means that $n + 2g - 2 > 0$. The Gromov-Witten invariants are

$$\langle \phi_1 \otimes \dots \otimes \phi_n \rangle_{g,n,\beta} = \int_{\bar{M}_{g,n}} I_{g,n,\beta}(\phi_1 \otimes \dots \otimes \phi_n).$$

Intuitively, if we consider n cycles $Z_i \subset X$ and a cycle $B \subset \bar{M}_{g,n}$ then

$$\int_B I_{g,n,\beta}(PD^{-1}(Z_1) \otimes \cdots \otimes PD^{-1}(Z_n))$$

is the number of stable maps $f: (C, x_1, \dots, x_n) \rightarrow X$ such that $f(x_i) \in Z_i$ and $(C, x_1, \dots, x_n) \in B$ when this is finite, and zero otherwise.

The maps $I_{g,n,\beta}$ satisfy the axioms of a cohomological field theory. These include:

- (i) The identity axiom: consider the forgetful map $\pi_n: \bar{M}_{g,n+1} \rightarrow \bar{M}_{g,n}$. Then

$$I_{g,n+1,\beta}(\phi_1 \otimes \cdots \otimes \phi_n \otimes 1) = \pi_n^*(I_{g,n,\beta}(\phi_1 \otimes \cdots \otimes \phi_n)).$$

It means $\langle \phi_1 \otimes \cdots \otimes \phi_n \otimes 1 \rangle_{g,n+1,\beta} = 0$.

- (ii) The divisor axiom: if $\psi \in H^2(X, \mathbb{C})$ then

$$\pi_{n,*}(I_{g,n+1,\beta}(\phi_1 \otimes \cdots \otimes \phi_n \otimes \psi)) = \left(\int_\beta \psi \right) \cdot I_{g,n,\beta}(\phi_1 \otimes \cdots \otimes \phi_n).$$

- (iii) The (first) degeneration axiom: consider the embedding

$$f: \bar{M}_{g_1,n_1+1} \times \bar{M}_{g_2,n_2+1} \hookrightarrow \bar{M}_{g,n}, \quad g = g_1 + g_2, \quad n = n_1 + n_2,$$

obtained by gluing the two curves along the last points. Then

$$\begin{aligned} & f^*(I_{g,n,\beta}(\phi_1 \otimes \cdots \otimes \phi_n)) \\ &= \sum_{\beta_1 + \beta_2 = \beta} (I_{g_1,n_1+1} \otimes I_{g_2,n_2+1})(\phi_1 \otimes \cdots \otimes \phi_{n_1} \otimes \Delta \otimes \phi_{n_1+1} \otimes \cdots \otimes \phi_{n_1+n_2}), \end{aligned}$$

where $\Delta \in H^*(X, \mathbb{C})^{\otimes 2}$ is Poincaré dual to the diagonal in $X \times X$.

6. QUANTUM COHOMOLOGY

We can use the $g = 0$ Gromov-Witten invariants to define a Frobenius structure on an open subset $M \subset H^{2*}(X, \mathbb{C})$. The metric is exactly as above, and we take flat co-ordinates t_i as there. The prepotential is

$$F(\phi) = \sum_{\beta \in H_2(X, \mathbb{Z})} \sum_{n \geq 3} \frac{1}{n!} \langle \phi \otimes \cdots \otimes \phi \rangle_{0,n,\beta},$$

where $\phi = \sum_i t_i e_i \in M$. We don't know convergence in general³ so people talk about formal Frobenius manifolds but we shall ignore these issues. The terms with $\beta = 0$ vanish except for $n = 3$ when they reproduce the cubic prepotential for the classical cohomology. Note that

$$\frac{\partial^3 F}{\partial t_i \partial t_j \partial t_k}(\phi) = \sum_{\beta \in H_2(X, \mathbb{Z})} \sum_{n \geq 3} \frac{1}{(n-3)!} \langle e_i \otimes e_j \otimes e_k \otimes \phi \otimes \cdots \otimes \phi \rangle_{0,n,\beta},$$

³See Zinger, Some conjectures on the asymptotic behavior of Gromov-Witten invariants.

The small quantum cohomology locus in M is obtained by taking $\phi \in H^0(X, \mathbb{C}) \oplus H^2(X, \mathbb{C})$. Using the identity and divisor axioms we find that on this locus

$$\frac{\partial^3 F}{\partial t_i \partial t_j \partial t_k}(\phi) \sum_{\beta \in H_2(X, \mathbb{Z})} \sum_{n \geq 3} \frac{\left(\int_{\beta} \phi \right)^{n-3}}{(n-3)!} \langle e_i \otimes e_j \otimes e_k \rangle_{0,3,\beta} = \sum_{\beta \in H_2(X, \mathbb{Z})} \langle e_i \otimes e_j \otimes e_k \rangle_{0,3,\beta} \cdot e^{\int_{\beta} \phi}.$$

It only depends on 3-point genus zero stable maps.

Let us prove WDVV on the small quantum cohomology locus. We must show that

$$\sum_{\beta_1, \beta_2 \in H_2(X, \mathbb{Z})} \langle e_i \otimes e_j \otimes e_r \rangle_{0,3,\beta} \cdot g^{rs} \cdot \langle e_s \otimes e_k \otimes e_l \rangle_{0,3,\beta}$$

is unchanged if we cycle (i, j, k, l) . But

$$\sum_{r,s} g^{rs} e_r \otimes e_s = \Delta,$$

so by the degeneration axiom the above is equal to

$$\sum_{\beta \in H_2(X, \mathbb{Z})} \langle e_i \otimes e_j \otimes e_k \otimes e_l \rangle_{0,4,\beta}$$

pulled back via an embedding $f: \bar{M}_{0,3} \times \bar{M}_{0,3} \hookrightarrow \bar{M}_{0,4}$ which puts i, j on one component and k, l on the other. But $\bar{M}_{0,4} = \mathbb{P}^1$ and the boundary consists of three points corresponding to (12)(34), (13)(24) and (14)(23). Any two points on $\mathbb{P}^1$ are homologous which gives the result.

7. EULER VECTOR FIELD

A grading of charge D on a Frobenius algebra is a grading of the underlying algebra $A = \bigoplus_{d \in \mathbb{Z}} A_d$ such that for $u_i \in A_{d_i}$ the form $g(u_1, u_2) = 0$ unless $d_1 + d_2 = D$. E.g. when $A = H^{2*}(X, \mathbb{C})$ we take $H^{2d}(X, \mathbb{C})$ to have degree d . We now weirdly define a derivation $\partial: A \rightarrow A$ by setting $\partial(u) = (d-1) \cdot u$ if $u \in A_d$. This satisfies

$$g(\partial u, v) + g(u, \partial v) = (D-2)g(u, v), \quad \partial(u * v) = \partial(u) * v + u * \partial(v) + u * v.$$

Definition 7.1. An Euler vector field on a Frobenius manifold is a holomorphic section $E: \mathcal{O}_M \rightarrow T_M$ which satisfies $\nabla(\nabla(E)) = 0$, and satisfies

$$\mathcal{L}ie_E(g) = (2-D) \cdot g, \quad \mathcal{L}ie_E(*) = *,$$

where $\mathcal{L}ie_E$ denotes the Lie derivative, and $D \in \mathbb{C}$ is some fixed complex number.

The number $D \in \mathbb{C}$ is called the conformal dimension or charge of the Frobenius manifold. In flat co-ordinates the condition $\nabla(\nabla(E)) = 0$ is equivalent to

$$E = \sum_{i,j} a_{ij} t_i \frac{\partial}{\partial t_j} + \sum_j b_j \frac{\partial}{\partial t_j},$$

for constants $a_{ij}, b_j \in \mathbb{C}$. Roughly, the Frobenius structure is preserved by an action of $\mathbb{C}$ (or in some cases $\mathbb{C}^*$) with generating vector field E . This acts by affine transformations on the flat co-ordinates t_i .

Example 7.2. In the case of classical cohomology we take a homogeneous basis $e_1, \dots, e_n$ for $H^{2*}(X, \mathbb{C})$ with $e_i \in H^{2d_i}(X, \mathbb{C})$. A possible Euler vector field is then

$$E = \sum_i (1 - d_i) t_i \frac{\partial}{\partial t_i} + \sum r_i \frac{\partial}{\partial t_i},$$

for arbitrary constants $r_i \in \mathbb{C}$. Note that

$$\left[E, \frac{\partial}{\partial t_i} \right] = (d_i - 1) \frac{\partial}{\partial t_i}.$$

For quantum cohomology there is no freedom: we have to take $r_i \in \mathbb{C}$ so that

$$\sum_i r_i e_i = -c_1(\omega_X) \in H^2(X, \mathbb{C}).$$

8. EXAMPLES

Example 8.1. Take X to be a point. The quantum cohomology is a 1-dimensional Frobenius manifold with co-ordinate t_1 and

$$e = \frac{\partial}{\partial t_1}, \quad E = t \frac{\partial}{\partial t}, \quad F(t_1) = \frac{1}{6} t_1^3,$$

$$\frac{\partial}{\partial t_1} * \frac{\partial}{\partial t_1} = \frac{\partial}{\partial t_1}, \quad g_{11} = g \left(\frac{\partial}{\partial t_1}, \frac{\partial}{\partial t_1} \right) = 1.$$

Example 8.2. Take $X = \mathbb{P}^1$. The quantum cohomology has co-ordinates (t_1, t_2) and

$$e = \frac{\partial}{\partial t_1}, \quad E = t_1 \frac{\partial}{\partial t_1} + 2 \frac{\partial}{\partial t_2}, \quad F(t_1, t_2) = \frac{1}{2} t_1^2 t_2 + e^{t_2},$$

$$\frac{\partial}{\partial t_2} * \frac{\partial}{\partial t_2} = e^{t_2} \frac{\partial}{\partial t_1}, \quad g_{11} = g_{22} = 0, \quad g_{12} = 1.$$

Example 8.3. Take $X = \mathbb{P}^2$. The quantum cohomology has co-ordinates (t_1, t_2, t_3) and

$$e = \frac{\partial}{\partial t_1}, \quad E = t_1 \frac{\partial}{\partial t_1} + 3 \frac{\partial}{\partial t_2} - t_3 \frac{\partial}{\partial t_3},$$

$$F(t_1, t_2, t_3) = \frac{1}{2} t_1^2 t_3 + \frac{1}{2} t_1 t_2^2 + \sum_{k \geq 1} \frac{N_k}{(3k-1)!} e^{kt_2} t_3^{3k-1},$$

where N_k is the number of degree k rational curves on X passing through a generic set of $3k-1$ points. The WDVV equation completely determines these numbers! Restricting to small quantum cohomology $t_3 = 0$ loses this information.